

Arduino code

```
#include <AFMotor.h>
```

```
#include <SoftwareSerial.h>
```

```
#include <Servo.h>
```

```
// Bluetooth RX, TX
```

```
SoftwareSerial BT(2, 3);
```

```
// Only 2 Motors Used
```

```
AF_DCMotor motor1(1); // Left Motor
```

```
AF_DCMotor motor4(4); // Right Motor
```

```
// Servo Motor
```

```
Servo roboticArm;
```

```
char command;
```

```
void setup() {
```

```
  Serial.begin(9600);
```

```
  BT.begin(9600);
```

```
  // Motor Speed
```

```
  motor1.setSpeed(200);
```

```
  motor4.setSpeed(200);
```



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```
// Servo Setup

roboticArm.attach(9);

roboticArm.write(90);


stopMotors();


Serial.println("River Cleaning Robot Ready");
}


void loop() {

    if (BT.available()) {

        command = BT.read();

        Serial.println(command);


        // Forward

        if (command == 'F') {

            forward();

        }


        // Backward

        else if (command == 'B') {
```



```
backward();  
  
}  
  
// Left  
  
else if (command == 'L') {  
  
    left();  
  
}  
  
// Right  
  
else if (command == 'R') {  
  
    right();  
  
}  
  
// Stop  
  
else if (command == 'S') {  
  
    stopMotors();  
  
}  
  
// Arm Up  
  
else if (command == 'U') {  
  
    roboticArm.write(150);  
  
}  
  
// Arm Down  
  
else if (command == 'D') {
```



```
        roboticArm.write(30);  
    }  
}  
}
```

```
// Forward
```

```
void forward() {  
    motor1.run(FORWARD);  
    motor4.run(FORWARD);  
}
```

```
// Backward
```

```
void backward() {  
    motor1.run(BACKWARD);  
    motor4.run(BACKWARD);  
}
```

```
// Left Turn
```

```
void left() {  
    motor1.run(BACKWARD);  
    motor4.run(FORWARD);  
}
```

```
// Right Turn
```

```
void right() {
```



```
motor1.run(FORWARD);  
  
motor4.run(BACKWARD);  
  
}
```

```
// Stop Motors
```

```
void stopMotors() {  
  
    motor1.run(RELEASE);  
  
    motor4.run(RELEASE);  
  
}
```

